

ASSIGNMENT NO.1

Menu Driven Program

1. Set Date
2. Add Days
3. Add Months
4. Add Year
5. Display Date
6. Exits

Code:

Program.java

```
public class Program {
    public static void main(String[] args) {
        Date objDate = new Date(1, 1, 2026);
        int choice;

        do {
            System.out.println("***** Menu Driven Program *****");
            System.out.println("1.Set Date");
            System.out.println("2.Add Day");
            System.out.println("3.Add Months");
            System.out.println("4.Add Years");
            System.out.println("5.Display Date");
            System.out.println("6.Exit");

            System.out.println("Enter your choice:");
            choice = ConsoleInput.getInt();

            switch (choice) {
                case 1:
                    System.out.println("Enter the Day:");
                    int day = ConsoleInput.getInt() ;

                    System.out.println("Enter the Month:");
                    int month = ConsoleInput.getInt() ;

                    System.out.println("Enter the Year:");
                    int year = ConsoleInput.getInt() ;
```

```

        objDate.setDate(day,month,year);
        break;
    case 2:
        System.out.println("Enter number of Day:");
        int days = ConsoleInput.getInt() ;
        objDate.addDays(days);
        break;

    case 3:
        System.out.println("Enter number of Months:");
        int months = ConsoleInput.getInt() ;
        objDate.addMonths(months);
        break;

    case 4:
        System.out.println("Enter number of Year:");
        int years = ConsoleInput.getInt() ;
        objDate.addYears(years);
        break;
    case 5:
        objDate.display();
        break;
    case 6:
        System.out.println("Exit...");
        break;
    default:
        System.out.println("Invalid Choice");
        break;
    }
} while (choice != 6);
}
}

```

Date.java

Code:

```
public class Date {
    private int day;
    private int month;
    private int year;
    private int[] noofDays = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
    public Date(int dd, int mm, int yy) {
        setDate(dd, mm, yy);
    }

    // Setter / Mutator
    public void setDate(int dd, int mm, int yy) {
        // Check Year
        if (yy < 1950 || yy > 3000)
            year = 2026;
        else
            year = yy;

        // Check Month
        if (mm < 1 || mm > 12)
            month = 1;
        else
            month = mm;

        // Check Day
        if (dd < 1 || dd > daysInMonth(month, year))
            day = 1;
        else
            day = dd;
    }

    // Find number of days in a month
    private int daysInMonth(int month, int year) {
        // February
        if (month == 2) {
            // Leap year
            if (year % 400 == 0 || (year % 4 == 0 && year % 100 != 0)) {
                return 29;
            }
            else {
                return 28;
            }
        }
    }
}
```

```

    }

    // 30 days months
    if (month == 4 || month == 6 ||
        month == 9 || month == 11) {
        return 30;
    }

    // 31 days months
    return 31;
}

// Add Days
public void addDays(int days) {
    while (days > 0) {
        int remainingDays =
            daysInMonth(month, year) - day;
        if (days <= remainingDays) {
            day = day + days;
            days = 0;
        }
        else {
            days = days - (remainingDays + 1);
            day = 1;
            if (month == 12) {
                month = 1;
                year++;
            }
            else {
                month++;
            }
        }
    }
}

// Add Months
public void addMonths(int months) {
    month = month + months;
    while (month > 12) {
        month = month - 12;
        year++;
    }

    int maxDays = daysInMonth(month, year);

```

```
        if (day > maxDays) {
            day = maxDays;
        }
    }

    // Add Years
    public void addYears(int years) {
        year = year + years;
        int maxDays = daysInMonth(month, year);
        if (day > maxDays) {
            day = maxDays;
        }
    }

    // Display Date
    public void display() {
        System.out.println(day + "/" + month + "/" + year);
    }

    public int getDay() {
        return day;
    }

    public int getMonth() {
        return month;
    }

    public int getYear() {
        return year;
    }
}
```